

Notes: Fuhrer (AER, 2000)

Habit Formation in Consumption and Its Implications for Monetary-Policy Models

Contents

1	Big picture	3
2	Data and measurement	3
2.1	Macroeconomic sample	3
2.2	Hump-shaped response	4
2.3	Consumption and habit variables	4
2.4	Rule-of-thumb consumers	4
3	Model: habit formation in consumption	4
3.1	Period utility	4
3.2	Habit reference level	5
3.3	Euler equation intuition	5
3.4	Recursive marginal-utility term	5
4	Linearized consumption function	5
4.1	Linear approximation	5
4.2	Ex ante real interest rate	6
4.3	Rule-of-thumb extension	6
5	Estimation strategy	6
5.1	Vector autoregression and rational expectations	6
5.2	Likelihood and nested tests	7
6	Empirical results	7
6.1	Parameter estimates	7
6.2	Likelihood-ratio tests	7
6.3	Autocorrelation and impulse-response evidence	7
7	Monetary-policy model	8
7.1	Policy reaction function	8
7.2	Price specification	8
7.3	Nonconsumption output block	8
8	Policy simulations and implications	8
8.1	Disinflation experiment	8
8.2	Memory in the habit stock	9
9	Tables and figures	9
9.1	Figure 1 and Table 1: consumption response and estimates	9
9.2	Figures 2 and 3: habit model versus VAR	10
9.3	Figures 4 and 5: removing habit formation versus full monetary model	11
9.4	Figures 6–8: restrictions and disinflation simulations	12

10 Short summary	13
11 Conclusion	14
11.1 Core conclusion	14
11.2 Economic intuition	14
11.3 Takeaway	14

1 Big picture

- **Question.** Can a monetary-policy model built from optimizing behavior replicate the gradual, hump-shaped response of consumption and inflation to shocks?
- **Main friction.** Standard permanent-income or life-cycle consumption models make consumption jump immediately when agents receive news about lifetime resources. Aggregate data instead show delayed, hump-shaped responses.
- **Key modification.** The paper adds habit formation to household utility: current utility depends on current consumption relative to a slowly changing reference level of past consumption.
- **Core empirical result.** Habit formation is strongly supported by the aggregate data. The restriction of no habit formation is rejected with very high statistical confidence.
- **Rule-of-thumb consumers.** The model also allows a fraction of income to accrue to consumers whose consumption tracks current income. This rule-of-thumb channel is statistically significant but distinct from habit formation.
- **Why habit formation matters.** Habit formation gives households a motive to smooth *changes* in consumption, not just consumption levels. This produces the hump-shaped response to income and interest-rate shocks.
- **Monetary-policy implication.** When the habit-formation consumption equation is embedded in a model with a Taylor-style reaction function and a sticky-price inflation equation, the model better matches the joint dynamics of consumption, output, interest rates, and inflation.
- **Disinflation implication.** The habit-formation model produces a more gradual and realistic decline in inflation during a disinflation experiment. Models without habit formation front-load the adjustment too strongly.
- **Economic takeaway.** A monetary-policy model that wants to use optimizing microfoundations should include non-time-separable preferences or another mechanism that reproduces observed hump-shaped aggregate dynamics.

2 Data and measurement

2.1 Macroeconomic sample

- The main estimation sample is quarterly data from 1966:1 to 1995:4.
- The starting date is chosen because the federal funds rate consistently traded above the discount rate from this period onward, indicating its use as the main monetary-policy instrument.
- The baseline VAR includes:
 - the effective federal funds rate;
 - log per capita chain-weighted nondurable goods and services consumption;
 - log per capita real disposable income;
 - log per capita nonconsumption chain-weighted GDP;
 - inflation from the chain-type price index for consumption;
 - the Journal of Commerce industrial materials commodity price index.
- Consumption, income, and nonconsumption GDP are detrended using a segmented linear trend with a break in 1974.

Interpretation: The data are chosen to study the monetary transmission mechanism. The key empirical object is the dynamic response of consumption and inflation to income, interest-rate, and policy shocks.

2.2 Hump-shaped response

- Empirical VARs show that consumption and inflation respond gradually to shocks.
- The peak response often occurs several quarters after the shock rather than immediately.
- This is inconsistent with the simplest permanent-income model, which predicts an immediate jump in consumption in response to new information about lifetime resources.

Interpretation: The paper treats the hump-shaped response not as a minor detail but as a key diagnostic for whether a structural monetary-policy model captures true aggregate dynamics.

2.3 Consumption and habit variables

The representative consumer's current consumption is C_t . The habit or reference level is Z_t . The parameter γ measures the importance of the reference level in utility. The parameter ρ measures persistence in the reference level.

- If $\gamma = 0$, the reference level does not matter: there is no habit formation.
- If ρ is close to zero, the reference level is approximately last period's consumption.
- If ρ is close to one, the reference level has long memory.

Interpretation: Habit formation means consumers dislike abrupt movements in consumption relative to a reference level built from past consumption.

2.4 Rule-of-thumb consumers

The paper allows a fraction λ of total income to accrue to rule-of-thumb consumers. These consumers set current consumption equal to current income:

$$c_t = y_t. \quad (1)$$

Interpretation: Rule-of-thumb behavior generates immediate response to current income. Habit formation instead delays part of the response. The two mechanisms are therefore logically and empirically distinct.

3 Model: habit formation in consumption

3.1 Period utility

The basic period utility function is

$$U_t = \frac{1}{1-\sigma} \left(\frac{C_t}{Z_t^\gamma} \right)^{1-\sigma}. \quad (2)$$

The lifetime utility function is

$$U = U_t + \beta U_{t+1} + \beta^2 U_{t+2} + \dots. \quad (3)$$

- σ controls curvature of utility.
- γ controls how strongly the reference level enters current utility.
- β is the discount factor.

Interpretation: Utility is non-time-separable because current consumption affects future reference consumption. This creates a dynamic link between current choices and future marginal utility.

3.2 Habit reference level

The reference level evolves as

$$Z_t = \rho Z_{t-1} + (1 - \rho)C_{t-1}. \quad (4)$$

- $\rho = 0$ means the reference level is simply last period's consumption.
- $\rho > 0$ creates memory beyond one period.
- A higher Z_t raises the standard against which current consumption is evaluated.

Interpretation: The reference level makes consumption dynamics sluggish because households care about not only where consumption is today, but how it compares to a moving standard.

3.3 Euler equation intuition

The first-order condition compares the utility loss from reducing current consumption with the discounted utility gain from increasing future consumption. With habit formation, reducing current consumption also changes the future reference level.

The Euler equation in the paper can be summarized as

$$MU_t = \beta E_t[(1 + r_{t+1})MU_{t+1}], \quad (5)$$

where marginal utility includes both direct utility from current consumption and the indirect effect of current consumption on future habit stocks.

Interpretation: The marginal utility term is more complicated than in the standard permanent-income model because current consumption changes future habits.

3.4 Recursive marginal-utility term

The paper defines a recursive term P_t that summarizes the discounted future effect of current consumption on future utility through the habit stock:

$$P_t \equiv \beta \rho E_t P_{t+1} + \frac{U_{t+1} C_{t+1}}{\rho Z_{t+1}}. \quad (6)$$

Interpretation: This term is the formal representation of the fact that today's consumption influences tomorrow's reference level and therefore tomorrow's marginal utility.

4 Linearized consumption function

4.1 Linear approximation

For empirical work, the paper linearizes the first-order conditions and budget constraint around steady state. The resulting log-linear consumption-income relation has the general structure

$$c_t - y_t = E_t \sum_{j=1}^{\infty} \mu^j [\Delta y_{t+j} + a_1^*(p_{t+j+1} - p_{t+j}) + a_2^*(z_{t+j+1} - z_{t+j}) - \delta^* r_{t+j+1}]. \quad (7)$$

- μ is the discount factor applied to future income and marginal-utility terms.
- p_t captures forward-looking marginal effects from future consumption and habits.

- z_t is the log habit/reference level.
- r_t is an ex ante real interest rate.

Interpretation: The consumption-income ratio depends on expected future income growth, expected future habit dynamics, and expected real rates. Habit terms create smoother and more delayed consumption responses.

4.2 Ex ante real interest rate

The ex ante real rate is a discounted average of expected short-term real interest rates:

$$r_t = (1 - d) \sum_{i=0}^{\infty} d^i E_t(f_{t+i} - \pi_{t+i+1}), \quad (8)$$

where f_t is the federal funds rate and π_{t+1} is inflation.

Interpretation: The consumption equation is affected by long-horizon expected real rates, not only the current nominal policy rate.

4.3 Rule-of-thumb extension

To allow Campbell-Mankiw-style rule-of-thumb consumers, the log consumption-income equation becomes

$$c_t - y_t = (1 - \lambda) E_t \sum_{j=1}^{\infty} \mu^j [\Delta y_{t+j} + a_1^*(p_{t+j+1} - p_{t+j}) + a_2^*(z_{t+j+1} - z_{t+j}) - \delta^* r_{t+j+1}] + \varepsilon_{ct}. \quad (9)$$

- λ is the income share of rule-of-thumb consumers.
- When $\lambda = 0$, all consumers behave as forward-looking optimizing consumers.
- When $\lambda > 0$, some current income enters current consumption directly.

Interpretation: The model nests the standard permanent-income model, a model with rule-of-thumb consumers, and a model with habit formation.

5 Estimation strategy

5.1 Vector autoregression and rational expectations

The paper estimates the model using a VAR representation that generates forecasts of future income growth, consumption growth, inflation, and real interest rates. The structural model can be written as a linear rational-expectations system:

$$\sum_{i=-\tau}^0 H_i x_{t+i} + \sum_{i=1}^{\theta} H_i E_t(x_{t+i}) = \varepsilon_t. \quad (10)$$

Using the Anderson-Moore procedure, expectations are solved in terms of current and lagged observables:

$$E_t(x_{t+k}) = \sum_{i=-\tau}^{-1} B_i E_t(x_{t+k+i}), \quad k > 0. \quad (11)$$

Interpretation: This method lets Fuhrer impose all rational-expectations restrictions implied by the consumption model while retaining the VAR as the unrestricted benchmark.

5.2 Likelihood and nested tests

After solving out expectations, the model has decision rules of the form

$$\sum_{i=-\tau}^0 S_i x_{t+i} = \varepsilon_t. \quad (12)$$

The likelihood is based on the structural residual covariance matrix:

$$L = T \left(\log |J| - \frac{1}{2} \log |\hat{\Omega}| \right). \quad (13)$$

Interpretation: Because each structural model is nested in the unrestricted VAR, the paper can evaluate restrictions using likelihood-ratio tests.

6 Empirical results

6.1 Parameter estimates

The maximum-likelihood estimates show:

- habit formation is economically important, with γ estimated near 0.80;
- the reference level is close to last period's consumption, with ρ near zero;
- rule-of-thumb behavior is important, with λ around 0.26;
- the curvature parameter σ is larger than one;
- the forward-looking discount parameter μ is close to one.

Interpretation: The estimates imply both habit formation and rule-of-thumb behavior. The model is not simply a disguised permanent-income model.

6.2 Likelihood-ratio tests

The paper reports strong rejections of restricted alternatives:

- The no-habit restriction $\gamma = 0$ is overwhelmingly rejected.
- The no-rule-of-thumb restriction $\lambda = 0$ is also strongly rejected.
- The full habit-formation model's cross-equation restrictions are not rejected relative to the unrestricted VAR at conventional levels.

Interpretation: Habit formation and rule-of-thumb behavior both matter, but the key contribution is that the optimizing habit model can impose many restrictions without destroying the fit to macro dynamics.

6.3 Autocorrelation and impulse-response evidence

The figures show that:

- the permanent-income model produces an immediate jump response to income shocks;
- the habit-formation model produces a hump-shaped response similar to the VAR;
- the habit-formation model captures the autocorrelation structure between consumption, income, interest rates, and inflation;
- removing both habit formation and rule-of-thumb behavior substantially worsens the dynamic fit.

Interpretation: The model’s success is mainly dynamic: it matches the shape and timing of aggregate responses to shocks.

7 Monetary-policy model

7.1 Policy reaction function

The monetary-policy rule is a Taylor-style reaction function:

$$f_t = \left(1 - \sum_i \alpha_{f,i}\right) (\bar{r} + \bar{\pi}) + \sum_i \alpha_{f,i} f_{t-i} + \sum_j \alpha_{\pi,j} (\pi_{t-j} - \bar{\pi}) + \sum_k \alpha_{y,k} y_{t-k} + \varepsilon_{ft}. \quad (14)$$

- f_t is the federal funds rate.
- $\bar{r}$ is the equilibrium real interest rate.
- $\bar{\pi}$ is the inflation target.
- y_t is an output gap.

Interpretation: The reaction function identifies the systematic component of monetary policy.

7.2 Price specification

The sticky-price block is a Fuhrer-Moore-style contracting model, equivalent to a two-sided inflation specification:

$$\pi_t = \frac{1}{2} \sum_{i=1}^k (\pi_{t-i} + \pi_{t+i}) + \alpha y_t + \varepsilon_{pt}. \quad (15)$$

Interpretation: Inflation depends on both past and future inflation and on excess demand. The model therefore contains explicit expectations and nominal rigidity.

7.3 Nonconsumption output block

The nonconsumption component of GDP obeys an IS-style relation:

$$\tilde{y}_t^o = \omega \tilde{y}_{t-1}^o - \gamma_r (r_{t-1} - \bar{r}) + \varepsilon_{yt}. \quad (16)$$

Interpretation: The full monetary-policy model allows interest rates to affect broader GDP, not only nondurables and services consumption.

8 Policy simulations and implications

8.1 Disinflation experiment

The paper studies an unanticipated decline in the long-run inflation target from 3 percent to 0 percent.

- Without habit formation, real variables respond too quickly.
- With habit formation, consumption and inflation adjust more gradually.
- The gradual consumption response feeds into a more gradual inflation response through the price specification.

Interpretation: Habit formation helps the model reproduce the long and variable lags of monetary policy.

8.2 Memory in the habit stock

The paper also compares a short-memory habit reference level with a more persistent reference level.

- Short memory means ρ_z is close to zero.
- Longer memory means ρ_z is close to one.
- Longer memory makes consumption and inflation paths more persistent in the disinflation simulation.

Interpretation: The deeper the memory in the reference consumption stock, the more gradual the response to policy changes.

9 Tables and figures

9.1 Figure 1 and Table 1: consumption response and estimates

Read:

- Figure 1 compares consumption responses to a unit income shock for different habit strengths.
- With $\gamma = 0$, the permanent-income model responds immediately.
- Higher γ produces more delayed, hump-shaped responses.
- Table 1 reports maximum-likelihood and GMM estimates of the habit-formation consumption equation.
- The key estimates support $\gamma > 0$ and $\lambda > 0$.

Economic message: The combination of Figure 1 and Table 1 shows why habit formation is both visually and statistically important.

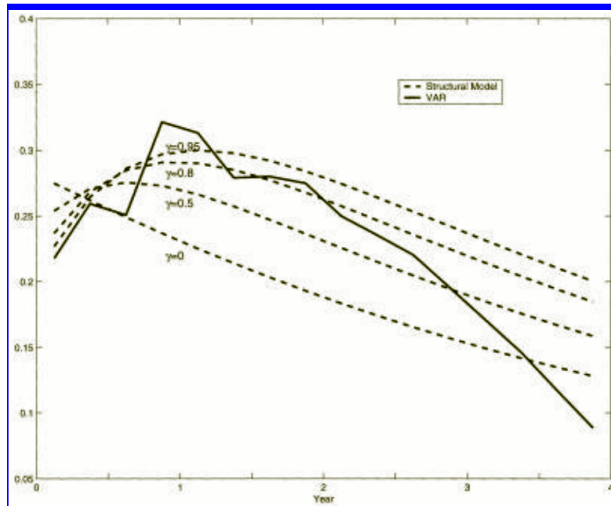


Figure 1: consumption response to an income shock

TABLE 1—ESTIMATION RESULTS

Coefficient	FIML estimates		GMM estimates	
	Estimate	Standard error	Estimate	Standard error
γ	0.80	0.19	0.90	1.83
ρ	0.0015	0.0039	0.052	7.41
λ	0.26	0.13	0.29	0.37
σ	6.11	1.81	13.02	7.85
μ	0.99	0.01	0.99	0.11
δ	28.49	5.17	94.89	8.00
Ljung-Box $Q(12)$	$(p\text{-value})$		Hansen's J -test:	
			4.99 (df = 11)	
Consumption	85.3	(0.00)	$p\text{-value} = 0.93$	
Income	10.7	(0.56)		
Funds rate	25.8	(0.01)		
Inflation	8.2	(0.77)		
Log-likelihood	2366.4			

Error correlations for consumption equation		
Lag	Autocorrelation	Partial correlation
1	0.53 (0.13)	0.53 (0.09)
2	0.37 (0.13)	0.12 (0.09)
3	0.40 (0.13)	0.23 (0.09)
4	0.14 (0.13)	-0.25 (0.09)

Asymptotic standard errors in parentheses

Table 1: estimation results

9.2 Figures 2 and 3: habit model versus VAR

Read:

- Figure 2 compares impulse responses from the VAR and the habit-formation model.
- The habit-formation model tracks the hump-shaped VAR responses reasonably well.
- Figure 3 compares the autocorrelation functions of the VAR and the habit model with rule-of-thumb consumers.
- The constrained model reproduces the key own- and cross-correlations among consumption, income, rates, and inflation.

Economic message: The habit model can match dynamic patterns that the standard permanent-income model misses.

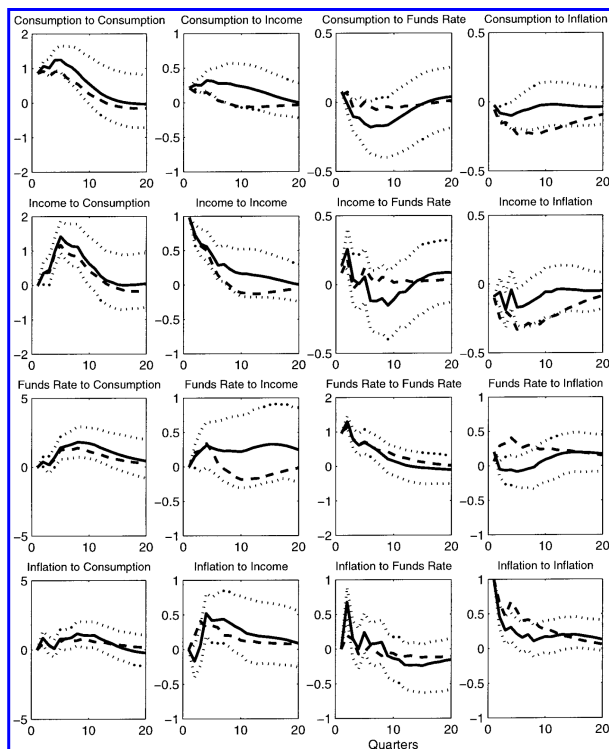


Figure 2: impulse responses

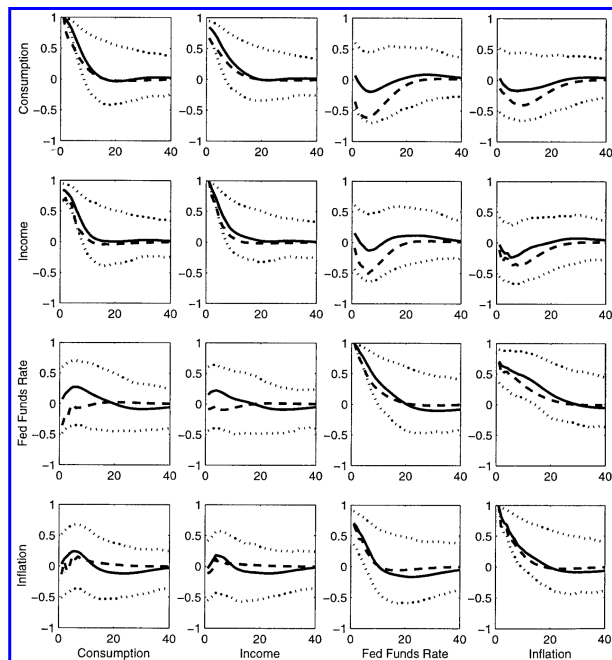


Figure 3: autocorrelation functions

9.3 Figures 4 and 5: removing habit formation versus full monetary model

Read:

- Figure 4 shows the pure life-cycle model with no habit formation and no rule-of-thumb consumers.
- The model fails to reproduce the main consumption dynamics and worsens inflation-consumption correlations.
- Figure 5 shows the habit-formation model with the reaction function and price specification imposed.
- The full model still closely resembles the unrestricted VAR autocorrelation function.

Economic message: The comparison isolates the contribution of habit formation. Habit formation materially improves the model without destroying the fit of the monetary-policy block.

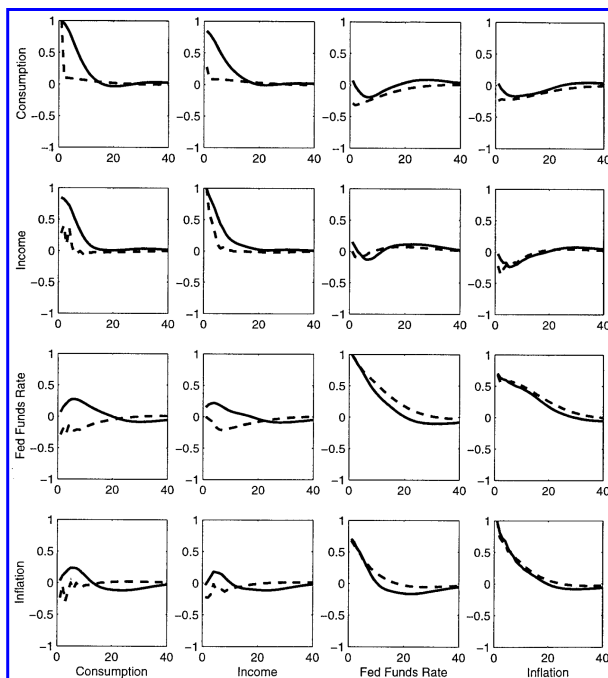


Figure 4: no habit formation or rule-of-thumb behavior

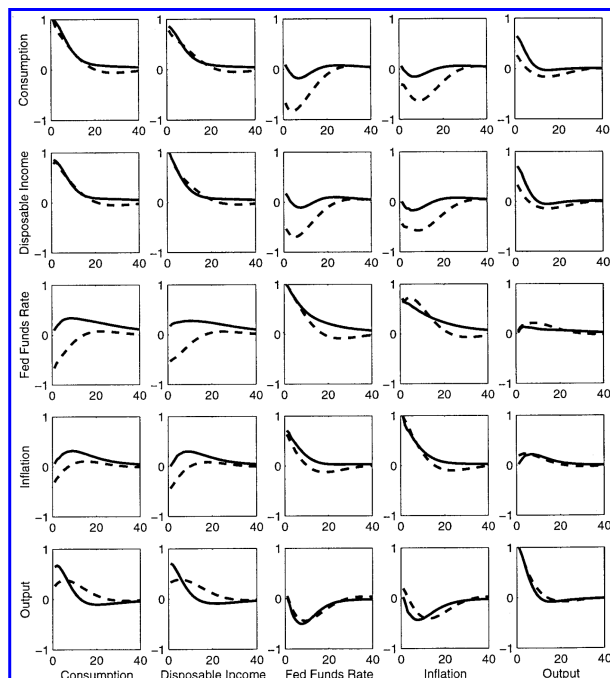


Figure 5: habit model with policy and price restrictions

9.4 Figures 6–8: restrictions and disinflation simulations

Read:

- Figure 6 checks the effect of imposing reaction-function and price-specification restrictions.
- Figure 7 compares a Fuhrer-style model with rule-of-thumb consumers to the habit-formation model during a disinflation.
- Figure 8 shows how the persistence of the reference level affects consumption and inflation paths.
- Longer habit memory produces more gradual adjustment.

Economic message: The monetary-policy payoff from habit formation is a more realistic gradual adjustment of spending and inflation after policy shocks.

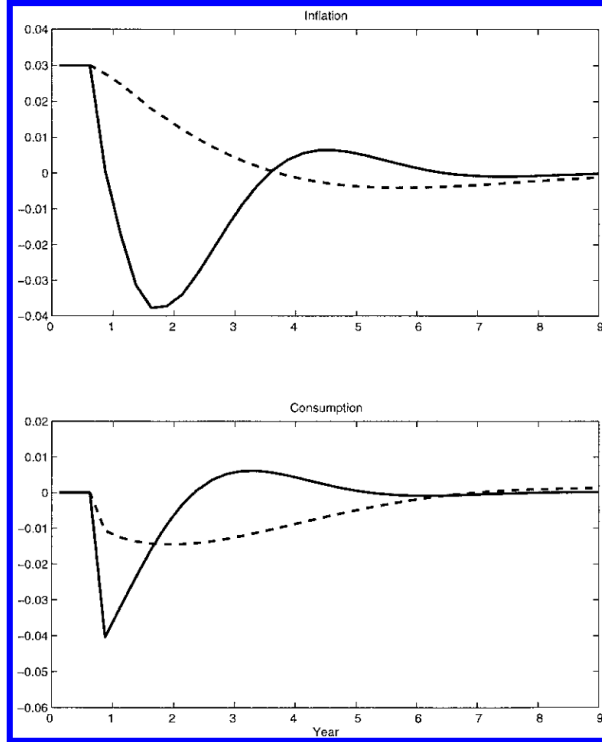
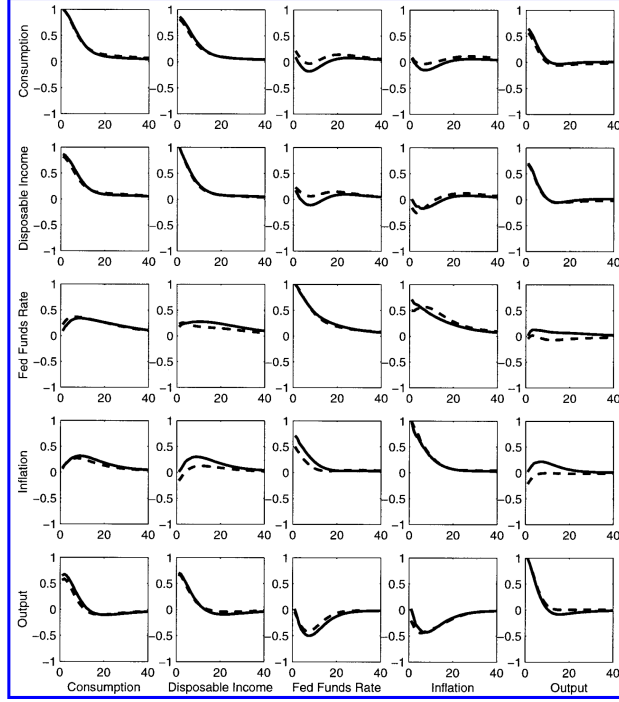


Figure 7: disinflation simulation

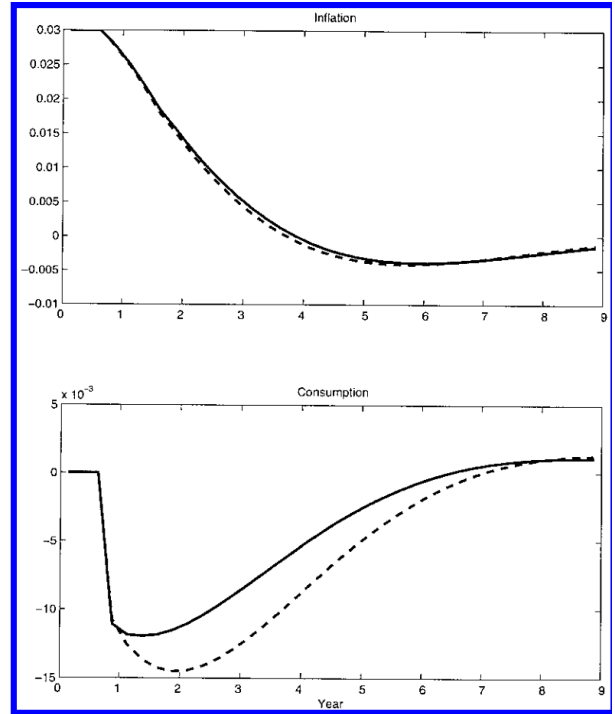


Figure 8: habit memory and disinflation paths

10 Short summary

- Standard permanent-income consumption models predict immediate jump responses to shocks.
- Aggregate consumption and inflation instead show delayed, hump-shaped dynamics.
- Fuhrer adds habit formation to consumer utility.

- Habit formation makes utility depend on current consumption relative to a reference level based on past consumption.
- The model also includes rule-of-thumb consumers, following Campbell and Mankiw.
- Maximum-likelihood estimates strongly support habit formation and rule-of-thumb behavior.
- The habit model matches VAR impulse responses and autocorrelation functions better than the standard life-cycle model.
- Embedding the habit consumption block in a monetary-policy model improves spending and inflation dynamics.
- In disinflation simulations, habit formation produces a more gradual decline in inflation.
- The paper argues that monetary-policy models should incorporate realistic short-run dynamics as well as optimizing foundations.

11 Conclusion

11.1 Core conclusion

Fuhrer shows that habit formation in consumption is both statistically significant and economically important for monetary-policy models. The standard permanent-income model misses a central feature of the data: consumption responds gradually and in a hump-shaped manner to shocks. Habit formation gives consumers a motive to smooth changes in consumption, allowing the model to replicate this pattern.

11.2 Economic intuition

When consumers care about consumption relative to past consumption, they do not immediately jump to a new consumption level after a shock. They adjust gradually because abrupt changes alter the habit stock and future marginal utility. This gradual consumption response feeds into inflation dynamics through the monetary-policy model, producing more realistic responses to policy actions.

11.3 Takeaway

The paper's main lesson is that microfounded monetary-policy models must be judged not only by whether they optimize, but also by whether they match the dynamic behavior of macroeconomic data. Habit formation provides a simple and empirically powerful way to reconcile optimizing behavior with the long and variable lags of monetary policy.